

Exercise 1: Prime numbers

Write an iterator that iterates over the first n prime numbers. Use this to print out the first 10,000 primes.

Exercise 2: Prime challenge

Write the function `primes(n)` that return a list with all prime numbers up to n

Use lambda functions and list comprehensions.

Try your best to complete it with as little code as possible

Exercise 3: List comprehensions

Let $i, j = 1, \dots, n$

(a) Generate a list with elements $[i, j]$.

(b) Generate a list with elements $[i, j]$ with $i < j$

(c) Generate a list with elements $i + j$ with both i and j prime and $i > j$.